

JEE ADVANCED - 8 | PAPER - 2

JEE 2022

Date 13th February, 2022	Maximum Marks 180	Duration 3 Hours
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General Instructions

- The question paper consists of 3 Subject (Subject I: **Physics**, Subject II: **Chemistry**, Subject III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
- Section 1** contains **2 types** of questions [**Type A and Type B**].
Type A contains **SIX (06) Multiple Correct Answers Type Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.
Type B contains **THREE (03) paragraphs**. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- Section 2** contains **6 Numerical Value Type Questions**. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
- For answering a question, an ANSWER SHEET (OMR SHEET) is provided separately. Please fill your **Test Code**, **Roll No.** and **Group** properly in the space given in the ANSWER SHEET.

Name of the Candidate (In CAPITALS) :

Roll Number :

OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

MARKING SCHEME

SECTION-1 | Type A

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

<i>Full Marks</i>	: +4 If only (all) the correct option(s) is(are) chosen;
<i>Partial Marks</i>	: +3 If all the four options are correct but ONLY three options are chosen;
<i>Partial Marks</i>	: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
<i>Partial Marks</i>	: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
<i>Zero Marks</i>	: 0 If unanswered;
<i>Negative Marks</i>	: -2 In all other cases.
- For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then
 - choosing ONLY (A), (B) and (D) will get +4 marks;
 - choosing ONLY (A) and (B) will get +2 marks;
 - choosing ONLY (A) and (D) will get +2 marks;
 - choosing ONLY (B) and (D) will get +2 marks;
 - choosing ONLY (A) will get +1 mark;
 - choosing ONLY (B) will get +1 mark;
 - choosing ONLY (D) will get +1 mark;
 - choosing no option(s) (i.e. the question is unanswered) will get 0 marks and
 - choosing any other option(s) will get -2 marks.

SECTION-1 | Type B

- This section contains **THREE (03) paragraphs**. Based on each paragraph, there are **TWO (02)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

<i>Full Marks</i>	: +3 If ONLY the correct option is chosen;
<i>Zero Marks</i>	: 0 If none of the options is chosen (i.e. the question is unanswered);
<i>Negative Marks</i>	: -1 In all other cases.

SECTION - 2

- This section contains **6 Integer Type Questions**. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks:	+3 If ONLY the correct Integer value is entered. There is NO negative marking .
Zero Marks:	0 In all other cases.

SUBJECT I : PHYSICS**60 MARKS****SECTION-1 | Type A**

This section consists of 6 Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

- A sound source which emits sound of frequency 10^3 Hz is kept on the ground. A detector is released from rest from a sufficient height above the source. Take $g = 10 \text{ m/s}^2$. It is observed that the frequency observed by the detector after 30s is double the initial frequency. Choose the correct statement(s):

(A) The speed of sound in the medium is 300 m/s

(B) The speed of sound in the medium is 330 m/s

(C) The graph of frequency observed by the detector with time is a straight line

(D) The graph of frequency observed by the detector with time is a parabola
- Consider a region where both uniform electric and magnetic fields E and B are present both along the z -axis. A positively charged particle of charge q and mass m is projected from the origin with initial velocity $V_0 \hat{i}$. Which of the following options are correct?

(A) The y -coordinate of the particle at time $t = \frac{\pi m}{qB}$ is $\frac{-2mV_0}{qB}$

(B) The distances between two consecutive points on the z -axis where the particle meets the z -axis are in ratio of $1:3:5$ and so on.

(C) The distances between two consecutive points on the z -axis where the particle meets the z -axis are in ratio of $1:4:9$ and so on

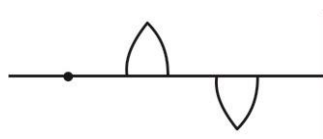
(D) The maximum x -coordinate the particle is $\frac{2mV_0}{qB}$
- The radiation emitted when an electron jumps from $n = 3$ to $n = 2$ orbit in a hydrogen atom, falls on a metal to produce photoelectrons. The electrons from the metal surface with the maximum kinetic energy are made to move perpendicular to a magnetic field of $3 \times 10^{-4} \text{ T}$ in radius of 10 mm . Then

(A) The kinetic energy of the electrons is 0.8 eV

(B) The kinetic energy of the electron is 0.16 eV

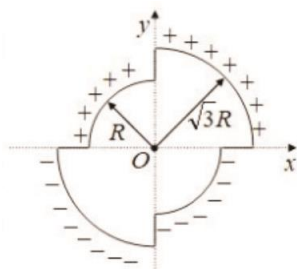
(C) The work function of the metal is 1.09 eV

(D) The work function of the metal is 2 eV
- An object is placed 100 cm from the screen. A convex lens of focal length 16 cm is placed between the object and the screen so as to obtain a sharp image on the screen. Now the lens is cut into two pieces along the principal axis and the bottom half is moved towards the screen so as to again obtain a sharp image. Choose the correct statement(s):



- (A) The original distance of the lens from the object is 20 cm
- (B) The distance of the bottom half of the lens is 40 cm from object
- (C) The distance of the bottom half of the lens is 80 cm from object
- (D) The images are formed on the opposite sides of the principal axis.

5. In the figure there are four arcs carrying positive or negative charge all of them have same magnitude of charge density λ . Choose the correct statement(s).



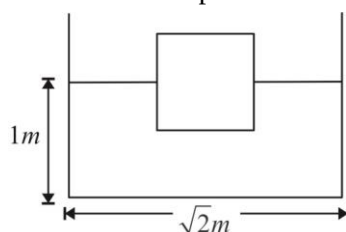
- (A) The net dipole moment for the given charge distribution is $4\sqrt{5}\lambda R^2$
 (B) The resultant electric field at the center and at any point on the axis passing through point O and perpendicular to the plane of arcs are opposite in direction
 (C) If a uniform electric field is switched on perpendicular to the plane then the charge distribution starts rotating about x-axis
 (D) Potential at the center of the given charge distribution is zero
6. In a Young's double slit experiment, distance between the slits and the screen is $1m$ and distance between the two slits is $5mm$. Light of wavelength $500nm$ is used. If slabs of thickness $2mm$ and $1.5mm$ having refractive index 1.5 and 1.4 respectively are placed in front of the first and the second slit respectively, then choose the correct statement(s):
- (A) Shift in the central maxima will be $8cm$
 (B) Shift is independent of the wavelength of light used
 (C) Shift will be in the direction of first slit
 (D) Shift will be in the direction of second slit

SECTION-1 | Type B

This section consists of THREE (03) paragraphs. Based on each paragraph, there are TWO (02) questions. Each question has FOUR options (A), (B), (C) and (D). ONLY ONE of these four options is the correct answer.

Paragraph for Question 7 - 8

A cube of wood of side $1m$ and density $500kg/m^3$ is dipped in a tank of water (density $1000 kg/m^3$). We want to push the cube to the bottom of the tank. The depth of the water in tank is $1m$, as shown figure. The cross section of the base of tank is $\sqrt{2}m \times \sqrt{2}m$ and the water does not spill out as the cube is pushed down.



7. As the cube is pushed down in the tank:
- (A) The force required to push it down increases continuously.
 (B) The force required to push it down decreases continuously.
 (C) The force required to push it down increases and then becomes constant.
 (D) The force required to push it down decreases and then becomes constant.
8. The minimum work required to be done by the person pushing the cube to the base of the tank is:
- (A) $1250J$ (B) $2500J$ (C) $1875J$ (D) $625J$

Paragraph for Question 9 - 10

A charge Q is uniformly distributed over a ring of radius R and mass m . It is kept on a rough horizontal surface with coefficient of friction $\mu = \frac{QB_0 R}{mg}$ with the plane of the ring parallel to the surface. At $t = 0$, a uniform vertical magnetic field $B = B_0 t^2$ is switched on.

9. Find the time at which ring will just start rotating about an axis perpendicular to its plane and passing through its center.

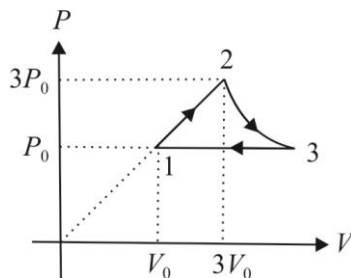
(A) $\frac{1}{\sqrt{3}}s$ (B) $\frac{1}{2}s$ (C) $1s$ (D) $2s$

10. Find the angular velocity of the ring at $t = 3s$.

(A) $\frac{QB_0}{2m}$ (B) $\frac{QB}{m}$ (C) $\frac{4QB_0}{3}$ (D) $\frac{2QB_0}{m}$

Paragraph for Question 11 - 12

One mole of an ideal monoatomic gas undergoes a cyclic process as shown in figure. Temperature at point 1 is $300K$ and process 2–3 is isothermal.



11. Heat capacity of process $1 \rightarrow 2$ is:

(A) $\frac{R}{2}$ (B) $\frac{2R}{2}$ (C) $\frac{5R}{2}$ (D) $2R$

12. The efficiency of cycle is:

(A) $\left(\frac{9\ln 3+4}{9\ln 3-4}\right)$ (B) $\left(\frac{9\ln 3-4}{9\ln 3+12}\right)$ (C) $\left(\frac{9\ln 3-4}{9\ln 3+16}\right)$ (D) $\left(\frac{9\ln 3-12}{9\ln 3+16}\right)$

SECTION-2

This section consists of 6 Numerical Value Type Questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08).

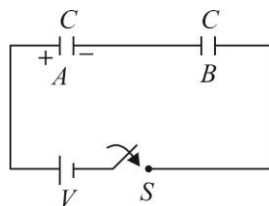
1. In a certain series LCR AC circuit, it is found that $X_L = 2X_C$ and phase difference between the current and voltage is $\pi/4$. If now capacitance is made $\frac{1}{4}th$ and the inductance & resistance are doubled then the phase difference between voltage and current becomes $\tan^{-1}\left(\frac{x\pi}{4}\right)$. Find the value of x .

2. A hollow steel sphere, weighing 200 kg is floating on water. A weight of 10 kg is to be placed on it in order to just submerge it at a temperature of 20°C . Find the weight (in gram rounded off to nearest integer) that should be decreased when temperature increases to 25°C so that the sphere keeps floating fully submerged.

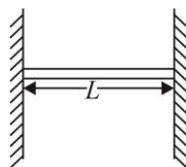
$$(\gamma_{\text{water}} = 1.5 \times 10^{-4} / ^\circ\text{C}, \alpha_{\text{steel}} = 1 \times 10^{-5} / ^\circ\text{C})$$

3. Initially S is open and capacitor A is charged up to potential difference $\frac{V}{2}$ and capacitor B is uncharged.

After closing the switch S the heat loss in the circuit is $\frac{CV^2}{2 \times n} J$. Find n .



4. Two radioactive samples A_1 and A_2 having half life 3 years and 2 years respectively have been decaying for many years. Today the number of atoms in the sample A_1 is twice the number of atoms in the sample A_2 . Both the samples had same number of atoms X years ago, calculate X .
5. A uniform rod of length 8 m is made of a material having Young's modulus $2 \times 10^{11} \text{ N/m}^2$, density 50 g/cm^3 and linear coefficient of expansion $2 \times 10^{-6} / ^\circ\text{C}$. The rod is placed between two vertical supports such that the rod just fits between the supports as shown in the figure. The coefficient of friction between the rod and left wall is $\mu = 0.2$ and between the rod and the right wall is also μ . Find the minimum rise in temperature (in $^\circ\text{C}$) so that the rod is in equilibrium. (Take $g = 10 \text{ m/s}^2$)



6. An electric dipole (dipole moment $= p$) is placed in a uniform electric field in stable equilibrium position at rest. Now it is rotated by a small angle and released. The time after which it comes to the equilibrium position again (for first time) is t . If the moment of inertia of the dipole about the axis of rotation is $x \frac{t^2 p E}{\pi^2}$, then find the value of x .

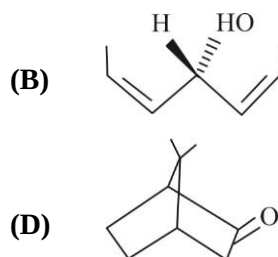
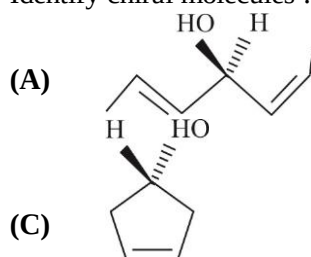
SUBJECT II : CHEMISTRY**60 MARKS****SECTION-1 | Type A**

This section consists of 6 Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

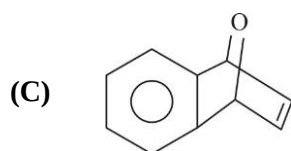
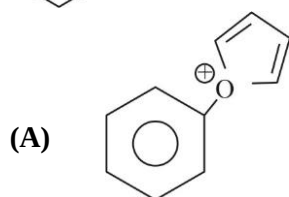
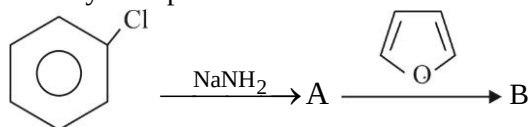
1. Choose the correct statements :

- (A) Fluorine can be prepared from fluorides by chemical oxidation using MnO_2
 (B) Anhydrous HCl is a good conductor of electricity than aq HCl
 (C) Dry chlorine does not bleach clothes
 (D) Colour of KI solution containing starch turns blue when chlorine water is added.

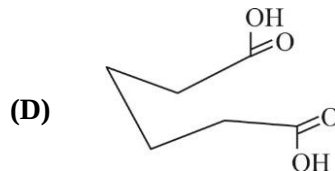
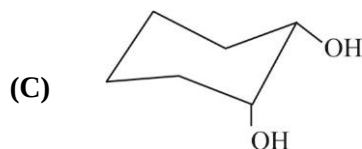
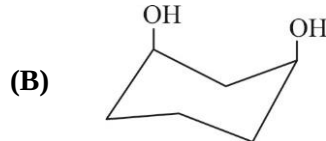
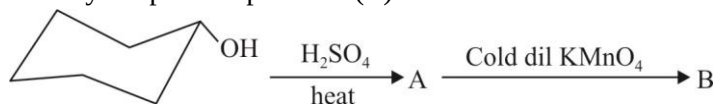
2. Identify chiral molecules :



3. Identify final product B.



4. Identify the product/products (B) :



5. Compare the reducing nature/power and mark the correct answer.

- (A) $\text{Li} > \text{Na} > \text{K} > \text{Rb} > \text{Cs}$ (B) $\text{Li} < \text{Na} < \text{K} < \text{Rb}$
 (C) $\text{Na} < \text{K} < \text{Rb} < \text{Li}$ (D) $\text{Be} < \text{Mg} < \text{Ca} < \text{Sr}$

6. Which of the following contain both ionic bond and pi bond?

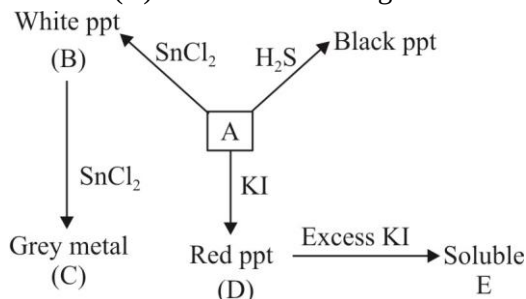
- (A) CaO (B) $\text{Ca}(\text{OH})_2$ (C) CaC_2 (D) CaCO_3

SECTION-1 | Type B

This section consists of THREE (03) paragraphs. Based on each paragraph, there are TWO (02) questions. Each question has FOUR options (A), (B), (C) and (D). ONLY ONE of these four options is the correct answer.

Paragraph for Question 7-8

A metal chloride (A) shows the following reactions



7. Compound A is :

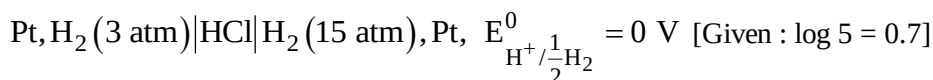
- (A) SnCl_4 (B) Hg_2Cl_2 (C) HgCl_2 (D) KCl

8. Red ppt (D) dissolves on addition of excess of KI and produce solution E. Identify E.

- (A) HgCl_2 (B) KHgI_4 (C) K_2HgI_4 (D) KCl

Paragraph for Question 9-10

Consider the following cell at 25°C



9. Calculate EMF of the following cells at 25°C :

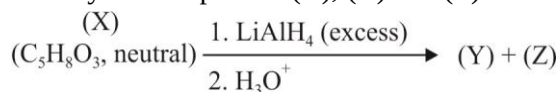
- (A) -0.2V (B) $+0.21\text{V}$ (C) -0.021V (D) $+0.021\text{V}$

10. The EMF of given cell can not be changed by :

- (A) Increasing the temperature (B) Increasing the pH of the solution
(C) By replacing H_2 gas to Cl_2 gas (D) Increasing the H_2 gas pressure at anode only.

Paragraph for Question 11-12

Identify the compounds (X), (Y) and (Z) in the following reaction



The compound (X) on hydrolysis yields two carboxylic acids. (Y) gives positive iodoform test.

11. Compound Y is :

- (A) $\text{C}_2\text{H}_5\text{OH}$ (B) CH_3OH
(C) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$ (D) $\text{CH}_3-\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$

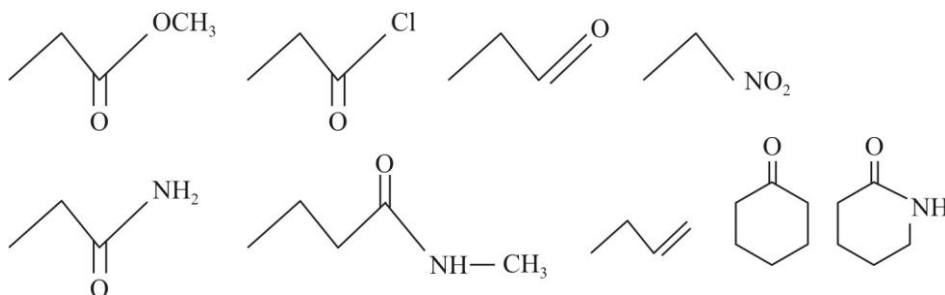
12. Compound Z is :

- (A) $\text{C}_2\text{H}_5\text{OH}$ (B) $\text{CH}_3-\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$
(C) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$ (D) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{OH}$

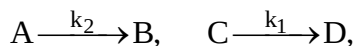
SECTION-2

This section consists of 6 Numerical Value Type Questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08).

1. NaBH_4 can be used to reduce several compounds. Number of compounds which can be reduced by NaBH_4 are:



2. In the following first order reactions :



the ratio of $\frac{k_1}{k_2}$, if at time t only 50% of A have been reacted whereas 96.875% of C has been reacted, is:

3. Pseudo first order rate for the reaction, $A + B \longrightarrow \text{Product}$, when studied in 0.1M of B is given by $-\frac{d[A]}{dt} = k[A]$, where, $k = 1.85 \times 10^{-4} \text{ sec}^{-1}$, rate of reaction of this second order reaction when $[A] = 0.01\text{M}$ and $[B] = 0.02\text{M}$ is $x \times 10^{-8}$. The value of x is ?

4. How many monochloro derivatives can be formed by monochlorination of

5. Consider following polymers

Polythene, PVC, Bakelite, Nylon-6,6, Buna-N, Buna-S, Teflon

Number of homopolymer = X

Number of copolymer polymer = Y

'XY' will be:

6. Calculate magnetic moment (BM) in potassium hexacyano manganate (II).

SUBJECT III : MATHEMATICS

60 MARKS

SECTION-1 | Type A

This section consists of 6 Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

- The solutions of the equation $9 \cos^{12} x + \cos^2 2x + 1 = 6 \cos^6 x \cos 2x + 6 \cos^6 x - 2 \cos 2x$ is/are

(A) $x = n\pi + \frac{\pi}{2}, n \in \mathbb{I}$ (B) $x = n\pi \pm \cos^{-1} \sqrt[4]{\frac{2}{3}}, n \in \mathbb{I}$

(C) $x = n\pi \pm \cos^{-1} \sqrt{\frac{2}{3}}, n \in \mathbb{I}$ (D) $x = n\pi, n \in \mathbb{I}$
- If a straight line through a point $P(\alpha, 2), \alpha \neq 0$, meets the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ at A and D and x - and y -axis at B and C respectively such that $PA \times PD = PB \times PC$, then ' α ' can lie in the interval

(A) $[7, \infty)$ (B) $(-12, -8)$ (C) $(-5, 0)$ (D) $(10, \infty)$
- Let $f(x) = \cos^{-1}(2x^2 - 1)$ then:

(A) $f(x)$ is continuous in $[-1, 1]$ (B) $f(x)$ is derivable in $(-1, 1)$

(C) range of $f(x)$ is $[0, \pi]$ (D) derivative of $f(x)$ w.r.t. $\sin^{-1} x$ at $x = \frac{1}{2}$ is 2
- A function $y = f(x)$ satisfying the differential equation $\frac{dy}{dx}(\sin x) - y \cos x + \frac{\sin^2 x}{x^2} = 0$ is such that, $y \rightarrow 0$ as $x \rightarrow \infty$ then the statement which is correct is:

(A) $\lim_{x \rightarrow 0} f(x) = 1$ (B) $\int_0^{\pi/2} f(x) dx$ is less than $\frac{\pi}{2}$

(C) $\int_0^{\pi/2} f(x) dx$ is greater than unity (D) $f(x)$ is an odd function
- Tangent drawn at point $P(1, 3)$ of a parabola intersects its tangent at vertex at $M(-1, 5)$ and cuts the axis of parabola at T . If $R(-5, 5)$ is point on SP ; where S is focus of the parabola, then :

(A) Slope of axis is -3 (B) Radius of circumcircle of ΔSMP is $\sqrt{\frac{5}{2}}$ units

(C) $(ST)^2 - (SM)^2 = (PM)^2$ (D) Tangent cuts the axis of parabola at $T(-3, 7)$
- Suppose that $f(x)$ is a differentiable invertible function, $f(x) \neq 0$ and $h(x) = \int_1^x f(t) dt$. Given that $f(1) = f'(1) = 1$ and $g(x)$ is inverse of $f(x)$.,
Let $G(x) = x^2 g(x) - x h(g(x)) \forall x \in \mathbb{R}$. Which of the following are correct.

(A) $G'(1) = 2$ (B) $G'(1) = 3$ (C) $G''(1) = 2$ (D) $G''(1) = 3$

SECTION-1 | Type B

This section consists of THREE (03) paragraphs. Based on each paragraph, there are TWO (02) questions. Each question has FOUR options (A), (B), (C) and (D). ONLY ONE of these four options is the correct answer.

Paragraph for question 7 - 8

Two tangents PA and PB are drawn from a point $P(h, k)$ to the ellipse $E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 (a > b)$. Angle of the tangents with the positive x-axis are θ_1 and θ_2 . Normals at A and B are intersecting at Q point. On the basis of above information answer the following questions.

7. Locus of P , if $\tan \theta_1 \cdot \tan \theta_2 = 4$, is :

(A) $\frac{y-b}{x-a} = 2(x+a)$

(B) $y^2 - b^2 = 2(x^2 + a^2)$

(C) $\frac{y+b}{x+a} = \frac{4(x-a)}{y-b}$

(D) $\frac{y+b}{x-b} = \frac{x+a}{y-b}$

8. Locus of P , if $\cot \theta_1 + \cot \theta_2 = \lambda$, is :

(A) $2xy = \lambda(y^2 - b^2)$

(B) $2xy - \lambda(b^2 - y^2) = 0$

(C) $xy = \lambda$

(D) $x^2 + xy = \lambda$

Paragraph for question 9 - 10

Let $f(x) = \begin{cases} xg\left(\frac{1}{x}\right), & x < 0 \\ x + ax^2 - x^2, & x \geq 0 \end{cases}$, where $g(t) = \lim_{x \rightarrow 0} (1 + a \tan x)^{t/x}$, then

9. If a is even prime natural number, then $g(2) =$

(A) e^2

(B) e^3

(C) e^4

(D) none of these

10. Set of all values of a for which function $f(x)$ is continuous at $x = 0$:

(A) $(-1, 10)$

(B) $(-\infty, \infty)$

(C) $[0, \infty)$

(D) none of these

Paragraph for question 11 - 12

Let $g(t) = \int_{x_1}^{x_2} f(t, x) dx$. Then $g'(t) = \int_{x_1}^{x_2} \frac{\partial}{\partial t} (f(t, x)) dx$. Consider $f(x) = \int_0^\pi \frac{\ln(1 + x \cos \theta)}{\cos \theta} d\theta$.

11. Range of $f(x)$ is :

(A) $(0, \pi)$

(B) $(0, \pi^2)$

(C) $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$

(D) $\left(\frac{-\pi^2}{2}, \frac{\pi^2}{2}\right)$

12. The number of critical points of $f(x)$, in $(-1, 1)$ is :

(A) 0

(B) 1

(C) 2

(D) infinitely many

SECTION-2

This section consists of 6 Numerical Value Type Questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08).

- The straight line $\frac{x}{4} + \frac{y}{3} = 1$ intersects the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ at two points A and B , there is a point P on this ellipse such that the area of ΔPAB is equal to $6(\sqrt{2}-1)$, the number of such points (P) is :
- Suppose that AB is the chord of contact of origin with respect to the circle $x^2 + y^2 - 4ax - 2ay + 4a^2 = 0$ ($a > 0$) such that $AB = 3$. Then the value of $\frac{4a}{\sqrt{5}}$ is :
- A straight line cuts the x-axis at point $A(1,0)$, and y-axis at point B , such that $\angle OAB = \alpha$ ($\alpha > \frac{\pi}{4}$). C is middle point of AB and B' is a mirror image of point B with respect to line OC and C' is a mirror image of point C with respect to line BB' , then the ratio of the areas of triangles ABB' and $BB'C'$ is equal to:
- If number of points common between $\|x| - |y| = 1$ and $2a|x||y| + 1 = 2|x| + a|y|$ are 8 then value of $3a$ (where $a < 1$) is _____.
- $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-coplanar vectors, and $\vec{d}$ be a non zero vector, which is perpendicular to $(\vec{a} + \vec{b} + \vec{c})$. Now if $\vec{d} = \sin x(\vec{a} \times \vec{b}) + \cos y(\vec{b} \times \vec{c}) + 2(\vec{c} \times \vec{a})$, then minimum value of $x^2 + y^2$ is equal to $\frac{\lambda\pi^2}{4}$, where λ is :
- $T\left(t, \frac{t^2}{2}\right)$ ($t > 0$) be a point on the curve $C : y = \frac{x^2}{2}$. A circle S touches the curve C at the point T and the x-axis. Denote by $(x(t), y(t))$ the center of circle S . If $\lim_{r \rightarrow 0^+} \int_r^1 x(t)y(t)dt = \frac{\tan \frac{3\pi}{8}}{6a}$, then the value of a is :